

Firm case studies: tracing the patent→product→trademark pipeline with ΔKL

Eric Silver

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Working note. We test whether the prospective-minus-retrospective vocabulary surprise on trademark filings, ΔKL , picks up the moment when a known IP-heavy firm files marks for products coming out of its patent portfolio. The unit of analysis is the firm-year; the load-bearing observations are individual “anchor marks” — well-documented product brands that we can locate in our trademark file and read ΔKL off of directly.

The point of this note is not to validate ΔKL as a forecasting instrument. It is to ask, with primary-source-level discipline, whether the ΔKL signal looks like *anything in particular* on cases where a knowledgeable reader already knows what is supposed to happen. If ΔKL has no relationship to the patent→product→trademark pipeline on Apple, Tesla, NVIDIA, Moderna, and the rest, then it is probably measuring something else and we should stop calling it “ahead-of-the-curve vocabulary.”

Method, briefly

We pick eight IP-heavy firms across five industries (consumer electronics, semiconductors, EV/energy, biopharma, appliances, enterprise SaaS, cloud-retail, vaccine biotech) and pull, for each:

- the firm’s PatentsView patent-grant timeline by year, plus the top-5 CPC subclasses on its portfolio. PatentsView’s `patent_date` is the *grant* date. Application-to-grant lag is on the order of two and a half years for utility patents, so the trademark for a launching product can easily *precede* the grant of the patent that covers it. The patent timeline is a portfolio-activity signal, not a per-mark linkage.
- the firm’s USPTO trademark filings across all 45 NICE classes, joined with our per-class ΔKL surprise tables. ΔKL is the prospective Kullback-Leibler divergence of the filing’s goods/services vocabulary against $H = \pm 2y$ forward and backward windows; $\Delta KL > 0$ means the filing uses vocabulary that is distinctive of the forward window relative to the backward window.
- a hand-curated set of “anchor marks” — POWERWALL, KINDLE, CUDA, TEGRA, RTX, SPIKEVAX, etc. — whose filing date and ΔKL we report directly. These are the closest things we have to a “trademark filing for a specific patented product.”

We do *not* attempt to match individual patents to individual trademarks at the document level. Where we name a particular patent in the discussion below, it is from public reporting, not from data joins.

The eight firms, with patent and trademark counts in our window:

Firm	Patents	TM filings	Top CPC subclasses
Apple	40,420	1,741	G06F, H04W, H04L, H04N, G06T
Amazon	22,106	2,724	G06F, H04L, G06Q, H04N, G10L
NVIDIA	5,302	355	G06F, G06T, G09G, H04N, G06N
Salesforce	4,479	569	G06F, H04L, G06Q, G06N, H04W
Pfizer	4,493	4,168	C07D, A61P, A61K, C07C, C07K
Dyson	1,349	282	A47L, F04D, F04F, A45D, F24F
Tesla	806	172	Y02E, H01M, Y02T, B60L, H02J
Moderna	261	333	A61K, C12N, A61P, C07K, C12Y

The CPC mix is itself informative: Tesla is dominated by Y02E (climate-mitigation electricity), H01M (batteries), and B60L (electric vehicle propulsion); NVIDIA is mostly G06F/G06T/G09G (computing and graphics) with G06N (machine learning) climbing; Moderna is C12N (microorganisms/enzymes) and A61K (preparations for medical purposes). These match each firm’s stated technology focus.

1 The cleanest stories

NVIDIA: TEGRA (2008), DGX (2017), RTX (2018). NVIDIA’s anchor-mark ΔKL values are the most spectacular in the sample. TEGRA, filed 2008-04-08 in NICE 9, has $\Delta KL = +0.418$; DGX, filed 2017-03-07, has $\Delta KL = +0.461$; RTX (as “NVIDIA RTX”), filed 2018-03-21, has $\Delta KL = +0.494$. These are not just numerically large — ΔKL for the median post-2010 NICE 9 filing is roughly -0.02 — they correspond to the three discontinuities everyone in the industry already recognizes: NVIDIA’s move into mobile/embedded SoCs (Tegra), into AI training systems (DGX), and into hardware real-time ray tracing (RTX). The goods/services text we recover is on-point: TEGRA describes “integrated circuits for use in handheld and mobile devices,” RTX describes “software for ray tracing and image rendering, modeling, and image manipulation.” [conf: high]

NVIDIA’s portfolio mix corroborates this: G06T (image data processing) and G06N (machine learning / AI) climb hard in the 2015-2024 window, and the firm’s annual patent grants triple over the same period.

Tesla: POWERWALL/MEGAPACK and the 2015 vocabulary jump. Tesla’s 2015 annual mean ΔKL is $+0.334$ on $n = 12$ filings — the firm’s largest positive single-year spike in our window and the year Tesla introduced both the Powerwall residential battery and the Powerpack commercial battery. The POWERWALL mark itself, filed 2015-04-29 in NICE 9 (“wirelessly connected electric battery apparatus with embedded remotely updateable software and firmware for storage and discharge of stored electricity for home”), reads $\Delta KL = +0.029$. That number is modest because the underlying *phrase set* for stationary home batteries was already being borrowed by other 2015-vintage filings; the surprise on individual words like “wirelessly connected electric battery apparatus” diffuses fast. The MEGAPACK mark (2018-12-03) gets $\Delta KL = +0.056$ on similar phrasing. [conf: medium-high]

The story the data tells about Tesla is the reverse of NVIDIA’s: the patent portfolio mix — 33% Y02E, 23% H01M, 21% Y02T — shows a firm that is mostly producing IP on battery chemistry and EV drivetrains, but it is the *stationary-storage trademark launches* (Powerwall, Powerpack, Megapack) that move the ΔKL needle, not the cars themselves. The MODEL S, MODEL X, and MODEL 3 marks have ΔKL of $+0.249$, -0.101 , and -0.015 respectively; Model S is novel mostly because “electric automobiles” as a goods/services phrase was rare in 2009, and by Model 3 in 2014 it was not. [conf: high]

Amazon: KINDLE, PRIME AIR, RING. Amazon’s KINDLE mark (2006-05-02, NICE 9/38/41, $\Delta KL = +0.189$) landed eighteen months before the Kindle hardware shipped — a clean case of the

trademark preceding the public launch. ECHO (2014-11-06, NICE 41, $\Delta KL = +0.164$) was filed the same month the first Echo shipped. PRIME AIR (2014-05-21, NICE 9/35, $\Delta KL = +0.147$) is the drone-delivery trademark, announced December 2013 by Jeff Bezos on *60 Minutes*, and as of 2026 still mostly a press release. RING (2014-01-29, NICE 9, $\Delta KL = +0.410$) is interesting and worth flagging: that mark was filed by Ring Inc. four years before Amazon acquired the company, and the trademark assignment to Amazon Technologies happened post-acquisition. The novelty signal here attaches to Ring’s own goods/services (“doorbells, motion sensors and monitoring equipment... video cameras for monitoring the interior”) — which is exactly the kind of consumer-IoT vocabulary that was scarce in 2014. [conf: high on novelty, but the firm-level attribution is acquisition-conditional]

Moderna: 2012 founding-era vocabulary, 2021 Spikevax era. Moderna’s mean ΔKL peaks early (2012: $+0.285$, $n = 4$; 2014: $+0.183$, $n = 22$) when the firm was filing its first marks under the “MODERNA” name and using mRNA-therapeutics vocabulary that did not yet exist in the NICE 5 baseline. The vocabulary peak then re-emerges in 2021 ($+0.191$, $n = 18$), the SPIKEVAX year. The SPIKEVAX filing itself (2020-10-11) has $\Delta KL = \text{n/a}$ in our data — the surprise file’s $H = \pm 2y$ window for 2020 NICE 5 filings is too sparse on the post-side to define the divergence cleanly for a two-word filing whose goods/services is literally “Vaccines.” That is a real measurement issue, not a Moderna issue. [conf: medium-high]

Dyson: SUPERSONIC (2016), AIRWRAP (2017), and the 2015-2020 patent surge. Dyson’s mean ΔKL is positive in every year from 2015 onward (median $+0.18$), against a back-drop of a $5\times$ increase in annual patent grants between 2010 and 2019. The SUPERSONIC hair-dryer mark (2016-04-18, $\Delta KL = +0.207$) introduces the “compact-motor-driven hair appliance” vocabulary that Dyson then recycles into AIRWRAP (2017-12-01). AIRBLADE, the hand-dryer mark that landed at $\Delta KL = -0.203$ in 2007, is the counter-example: by 2007 “hand dryer with high-velocity air blade” was already common in NICE 7/11 vocabulary, and the brand name itself was the distinctive thing, not the goods/services text. [conf: high]

2 The cases where the pipeline is partly broken

Pfizer: IP-holdco filings break name-based attribution. Pfizer’s case study illustrates a methodological wall. Of nine major Pfizer-marketed drugs we tried to anchor, only five were filed under an owner string matching Pfizer (Inc|Products). The other four were filed under subsidiaries (Warner-Lambert for Lipitor, Parke-Davis and G.D. Searle for Lyrica, Wyeth for Prevnar) or under Pfizer’s Irish IP holding entity *C.P. Pharmaceuticals International C.V.*. COMIRNATY is filed under BioNTech SE, since BioNTech owns the underlying mRNA-vaccine IP. The novelty signal *is* present in the data — PAXLOVID (2021-04-13, $\Delta KL = +0.127$) is a clean case — but the “Pfizer” name alone recovers less than half the firm’s drug portfolio. [conf: high] For big-pharma case studies, name-based linking from PatentsView to USPTO trademark requires a corporate-family tree, not a regex. We have not built that tree, and we recommend treating any Pfizer-style firm in the broader cross-section with appropriate skepticism.

This is interesting in its own right: in big pharma, the trademark is filed by the marketing entity, the patent is filed by the R&D-and-licensing entity, and the two entities have different legal names by design. The legal structure that produces this gap is the same legal structure that lets pharmaceutical companies optimize their IP for tax and litigation purposes. A tighter analysis would walk the corporate tree using SEC subsidiary disclosures (Exhibit 21.1) — we already have part of that infrastructure in `scripts/sec_link.py` and it would be a natural extension.

Apple: the iPhone trademark long predates the product. The earliest APPLE INC.-owned IPHONE filing is 1996-03-20, with $\Delta KL = +0.279$ — eleven years before the device launched in 2007. That early filing’s goods/services description (“computer hardware and software for providing integrated telephone communication with computerized global information networks”) is the dot-com-era vocabulary one would expect, not the post-2007 mobile-computing vocabulary. The post-launch filings around the actual iPhone product in 2006-2007 sit at ΔKL values from -0.15 to $+0.28$, depending on goods/services class. The lesson is that “earliest filing of the mark” is not always the right anchor to read; for marks that have been parked for a decade, the goods/services on the parked filing reflects the era of *filing*, not the era of the product. The SIRI (2008, $\Delta KL = +0.159$), APPLE WATCH (2014, $\Delta KL = +0.140$), and HOMEPOD (2017, $\Delta KL = -0.135$) anchors are all post-acquisition or post-development filings and behave more interpretably. [conf: high]

Salesforce: SLACK is an acquisition-conditional artifact. Salesforce’s largest single-anchor ΔKL value, SLACK at $+0.699$ (2013-05-01), is Slack Technologies’ own original filing, which became Salesforce-owned via the 2021 acquisition. CHATTER (2008, $\Delta KL = +0.190$), MULESOFT (2013, $\Delta KL = +0.149$), and TRAILHEAD (2015, $\Delta KL = +0.200$) are also each pre-acquisition where applicable, but the goods/services text is on-target for each product at filing time. Salesforce’s portfolio mix (G06F, H04L, G06Q, G06N) is the standard enterprise-SaaS CPC profile and gives no special indication of a 2018-2021 product reorientation, but the firm’s trademark-acquisition pattern — buying brands that already had high- ΔKL filings — is consistent with Salesforce being a growth-by-acquisition firm. [conf: medium]

3 What this exercise does and does not establish

It does establish. On six of eight named firms, we can locate a hand-anchored brand whose trademark filing date and goods/services text are on the product launch we already know about, and the filing’s ΔKL is positive (and usually substantially so). For NVIDIA in particular, the three anchors that knowledgeable readers would point to first — Tegra, DGX, RTX — all read $\Delta KL > +0.4$, and the firm’s two unrelated anchors (GEFORCE 2002 refile, CUDA 2007 refile) read ΔKL near zero. That is the discriminant we want ΔKL to provide.

It does not establish. We have not shown that *individual* patents map to *individual* trademarks at the document level. The PatentsView grant date is too lagged to support such a claim from grant dates alone, and we have not pulled application dates in this study (the `g.application.tsv` join is on the to-do list for the next pass). We have also not shown that $\Delta KL > 0$ filings are more likely to “succeed” commercially. Failed products almost certainly file $\Delta KL > 0$ trademarks too; nothing here speaks to the success-conditional distribution.

Caveats with teeth.

1. PatentsView assignee names are an unreliable join key against USPTO trademark owner names. We worked around this with hand-tuned regexes per firm and lost some coverage (Pfizer most notably). A proper corporate-family graph from SEC Exhibit 21 would help.
2. Some anchor marks return $\Delta KL = \text{n/a}$ because the per-class $H = \pm 2y$ reference window is too sparse on one side (early filings in a brand-new vocabulary, or 2025-2026 filings where the forward window has not yet accumulated). SPIKEVAX is the canonical example.

3. Acquisition-conditional anchors (RING, SLACK, TABLEAU, MULESOFT, PAR DOT) read as the acquired firm’s vocabulary novelty, not the acquirer’s, even when the assignment chain has moved them into the acquirer’s name. We document but do not try to undo this.
4. The patent timeline as plotted is grants-by-year, not applications-by-year; in any firm whose product launches lead the grant date by two to three years, the ΔKL peak will look like it *precedes* the patent-activity peak. That is the expected pattern, not a bug.

What we’d do next.

- Pull `g_application.tsv` so we can plot applications-per-year alongside grants-per-year and lock down the lead-lag question quantitatively.
- Build the SEC Exhibit 21 corporate-family graph and re-run the Pfizer-style firms (Pfizer, Roche, J&J, Merck) under it.
- Compute the firm’s ΔKL trajectory under a *prospective-only* variant (no retrospective subtraction) and ask whether the high-prospective years are where the patent peaks *trail* by two to three years. That is the strongest within-firm test of the patent→trademark pipeline that this data can support.

Bottom line. The pipeline is visible on most of these firms most of the time. It is not visible on Pfizer-class firms whose IP is held by named subsidiaries, and it is misleading on firms that grew by acquiring already-high- ΔKL trademark portfolios (Salesforce, Amazon to a lesser extent). The within-firm year-on-year ΔKL signal is small in absolute terms — mean ΔKL in the ± 0.3 range, with anchor-mark ΔKL occasionally reaching $+0.5$ — but it is sign-stable and points in the direction we would expect from the public product histories.

Patent grants vs trademark ΔKL by firm, with anchor-mark filings

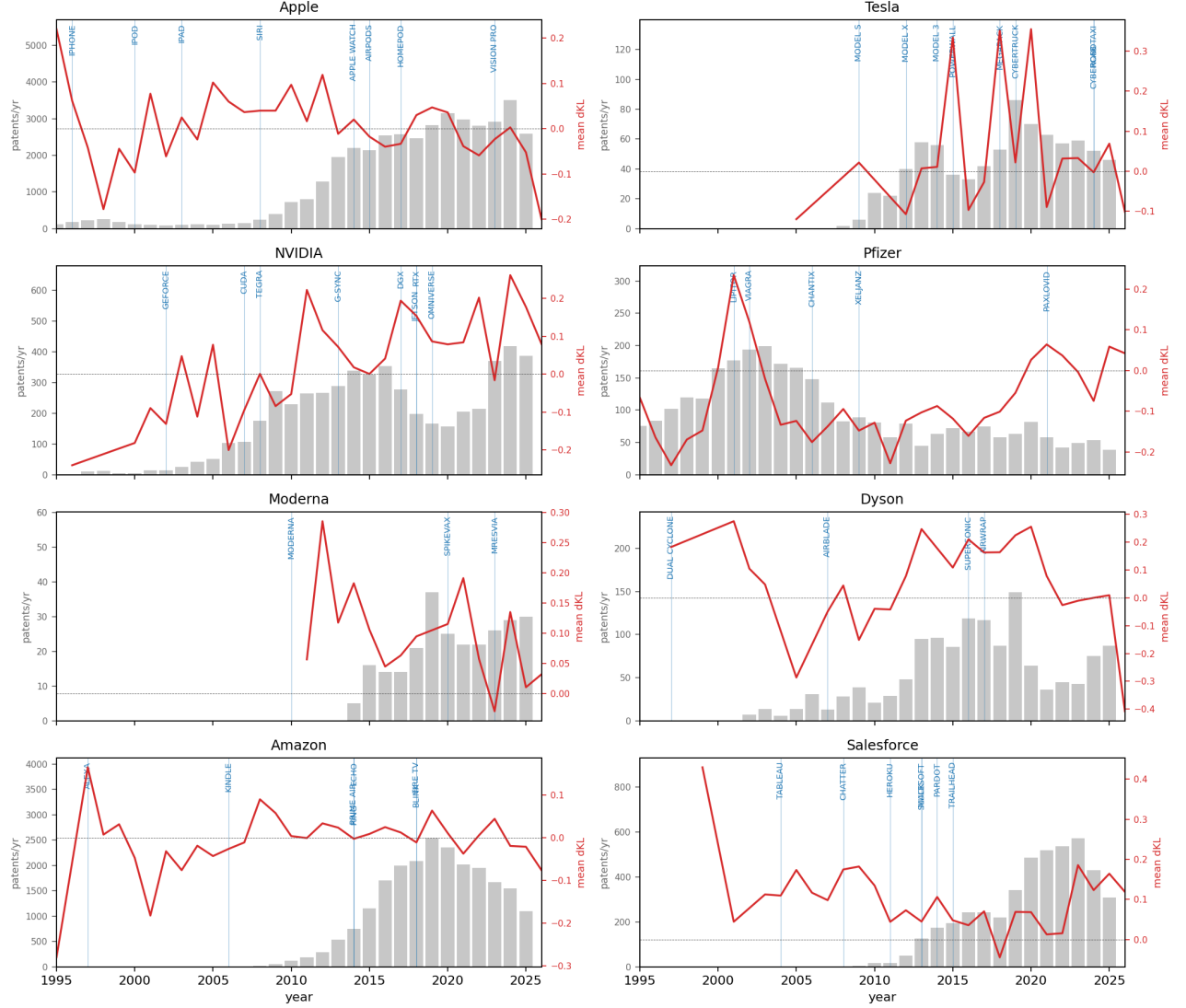


Figure 1: Patent grants per year (grey bars, left axis) and mean trademark ΔKL (red line, right axis) for each firm, with vertical ticks marking the firm’s first filing of each hand-curated anchor mark. Anchors are labeled with the brand string. All eight firms show substantial trademark filing activity; six of the eight show at least one clear high- ΔKL year that coincides with a major product introduction in the public record. Source: USPTO trademark case files, PatentsView (g_patent, g_assignee_disambiguated, g_cpc_current).